

James Fraser

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SUMMARY

Experienced researcher capable of bridging the gap between academic papers, data analytics, and production software. With a proven track record leading and engaging with multidisciplinary teams in healthcare, manufacturing, and education to resolve complex challenges through practical, leading-edge AI solutions. Actively developing environments agentic environment and looking to Current Technical Focus: Actively researching and developing Agentic AI orchestration and Agentic RAG architectures to produce self-evaluating and correcting pipelines to support complex client needs.

TECHNICAL COMPETENCIES

Python - Machine Learning and AI projects

- Machine Learning (Pandas/Polars, Numpy, Scikit-Learn, NLTK, Mediapipe, OpenCV)
- Web (FastAPI, Pydantic, SQLAlchemy)

JavaScript - (Web for PhD Project) NodeJS, Javascript, jQuery, React, AngularJS

C/C++ - Primarily used in game development and high-performance audiovisual performances.

Databases - MySQL, Redis, MongoDB

Docker, Git, Jenkins, CI/CD, RESTful, API, GIT, HTML, JSON, CSS

Java/JavaFx - Cad/CNC Software Project

PROFESSIONAL EXPERIENCE

Research Software Developer

Protocase

2024 - January 2026, Sydney, Nova Scotia

- One of two primary JavaFX developers for Protocase Designer and Protocase Designer that streamlines 3D modelling to custom manufacturing.
- Developed Industry leading free CNC capabilities with automated price quotes by extracting features from 3D models.
- Initial research for machine vision to automate feature extraction of complex geometrical data to reduce manual review and overhead.
- Consulted on the deployment of local AI tools, including automated meeting synthesis and sales call digitization using **WhisperX** and custom **local ChatAgents** to enhance internal productivity.

Postdoctoral Associate

Western University

2024-2025, London, Ontario

- Architected specialized researchers team in developing a pioneering healthcare application using AI to track patient progression for tele-rehabilitation.
- Patient bio-metrics tracked in real-time, using computer vision tools (Mediapipe) , to evaluate patient physical therapy progression.
- Clinical Decision Support System to collect data, progress-track continued physical therapy, and recommend surgical necessity.

Software Developer and Research Support II

Western University

2020-2023, London, Ontario

- Lead Developer for Research Team (2-3 programmers).
- Consulting with multidisciplinary teams of medical professionals, PhD graduates, developers, and researchers.
- Co-authored 12 research papers on the impact of COVID-19 on family workload, sexuality, anxiety, and depression - responsible for survey processing algorithms and analyses.
- Key Development Projects included:
 - Developed a cutting-edge Telehealth application to track patient's range of motion and evaluate physiotherapy progress.
 - Oculus Rift game to gamify physical therapy and evaluate patient rehabilitation with rotator cuff injuries.
 - Designed the processing platform for a virtual diagnostic tool for Carpal Tunnel Syndrome (which led to a PhD and 5 publications for the trainee).
 - Data analytics platform to extract COVID-19 patient data related to family workload, anxiety, and depression.

Lead Developer/Researcher

University of Guelph

2016-2018

- The architect of (IFS), a research and analytics platform designed to identify student learning profiles and to evaluate student learning.
- Managed and mentored a small team of junior software developers, including MSc students and co-ops.
- Utilized clustering and random forest technique to develop profiles for students to determine key factoring in learning environment.
- The experiment website ran for four years and supported five graduate degrees.

Research Software Developer

Christie Digital Systems

2013-2016

- Research and development of patented projection mapping prototype.
- Projection Mapping system presented at significant conferences (SIGGRAPH and ISE).
- Focused on machine vision techniques, 3D reconstruction, and projection mapping.
- Developed prototypes for facial and demographic recognition systems.
- Implemented advanced research algorithms for camera orientation in projection mapping.

Application Software, Software Developer

Christie Digital Systems

2012-2016

- Front-end development using C++/Qt/Web for digital cinema and visual solution projectors and touch panels.
- Team Scrum Master for Agile process and organized software development team sprints.
- Professional software development continuous integration build system, unit testing, code reviews, and constant QA validation.
- Developed a JSON RPC communication system for the company's projectors and client interfaces.
- Website development for smart canvas applications.
- Development iOS awarding-winning project remote applications.

Game Programmer

Big Blue Bubble

2008-2011, London, Ontario

- Lead Programmer: released 6 iPhone/iPad Games (Fighting Fantasy Series)
- Programmer: Developed two games for the Wii and 2 for PC.
- Developed tools for the company's PC division.
- Produced 2D and 3D video games.

EDUCATION

PhD in Computational Science

University of Guelph • Guelph, Ontario • 2023 • 4.0

- **2017:** Alexander Graham Bell Canada Graduate Scholarship (NSERC CGS-D, 3-Year) (University of Guelph)
- **2017:** College of Physical & Engineering Science Graduate Dean's Scholarship (University of Guelph)
- **2016:** Ontario Graduate Scholarship (OGS) (University of Guelph)
- **2016:** Achievement Scholarship Graduate Studies (University of Guelph)

MSc in Computer Science

University of Western Ontario • London, Ontario • 2012 • 4.0

- **2007:** Nomination for Graduate Teaching Assistant Award (Western University)
- **2007:** Achievement Scholarship Graduate Studies (Western University)
- **2006:** Achievement Scholarship Graduate Studies (Western University)

Honors Bachelor of Science: Computer Science

Saint Mary's University • Halifax, Nova Scotia • 2006 • 3.75

- **2006:** ACM 2nd Place Finish, APICS Programming Contest
- **2002-2004:** Dean's List for Academic Achievement (Saint Mary's University)

AWARDS & HONORS

Alexander Graham Bell Canada Graduate Scholarship

University of Guelph • 2017-2022

- **NSERC CGS-D is one of the largest research awards given in Canada to PhD Students.**

Ontario Graduate Scholarship (OGS)

University of Guelph • 2016-2018

- **OGS a significant award that is given throughout Ontario which covers your full tuition.**

Achievement Scholarship

University of Guelph • 2016

- Graduate Studies.

Achievement Scholarship

Western University • 2006-2008

- Graduate Studies.

Dean's List for Academic Achievement

